

From the Paper's Methods to an Analytic Dataset
SPPH 604 · Application of Advanced Epidemiological Methods

Companion to reproduction/COHORT_BUILD_STEPS.md · Kueh et
al. (2026), BMJ Open 16:e113719

This deck

Eight steps, from a CDC download to `masld_analytic.rds`

Every step answers the same four questions. That repetition is the point.

The question this deck answers

You have a published Methods section. You have CDC's website. **How do you get from one to the other, and how do you defend every choice you made on the way?**

- ▶ The paper: Kueh MTW et al., *BMJ Open* 2026;16:e113719 — body weight phenotypes and mortality in MASLD, NHANES 2007–2018
- ▶ The code: `reproduction/R/00_setup.R` → `01_download.R` → `02_build_analytic.R`
- ▶ The destination: **6,371 rows**, against a published **6,300**
*The gap is not a failure. **Being able to say why it exists is the skill.***

The one habit

A Methods sentence is **not** code. Between the sentence and the code there is always a decision — and in a published paper the decision is almost always invisible.

Important

Your job at P1 is not to guess the authors' decision. It is to **make your own, write down why, and count what it cost.**

Every step in this deck has the same four lines:

1. **What the paper says** — quoted, verbatim
2. **Component** → **variables** — what that sentence forces you to download
3. **The decision** — and *why*, because the sentence did not fully determine it
4. **N afterwards** — and the *type* of the loss

Three scripts, one of which matters

Script	What it decides	Writes
00_setup.R	6 cycles, 18 component names, folder paths. No analysis switch.	nothing
01_download.R	nothing analytic — fetches raw files, records a manifest	data/raw/, _download_manifest.csv
02_build_analytic.R	everything : merge → FLI → MASLD → groups → mortality → funnel	masld_analytic.rds, sample_funnel.csv

- ▶ Cycle letters used throughout: **E** 07-08 · **F** 09-10 · **G** 11-12 · **H** 13-14 · **I** 15-16 · **J** 17-18
- ▶ The *file* carries the suffix (BMX_H.xpt); the *variable* usually does not (BMXBMI in all six)

Where we are going: the funnel

Step	Criterion	N, full analytic file	Type
0	All NHANES 2007–2018 records	59,842	—
1	Age ≥ 18	36,580	ELIGIBILITY
2	FLI computable	14,989	DESIGN + MISSINGNESS
3	Hepatic steatosis (FLI ≥ 60)	6,376	TARGET POPULATION
4	MASLD (≥ 1 cardiometabolic criterion)	6,374	TARGET POPULATION
5	Phenotype group assignable	6,374	MISSINGNESS
6	Mortality-eligible and linked	6,371	OUTCOME ASCERTAINMENT
7	Alcohol exclusion applied =	6,371	SWITCH

Step 0 — What you download

Three downloads

#	What	Count
1	18 components $\times$ 6 cycles, from wwwn.cdc.gov/.../DataFiles/<COMP>_<suffix>.xpt	108 files
2	Multum drug lexicon RXQ_DRUG.xpt — tried at 2015, 2017, 2013	0 — all three 404
3	Linked mortality, ftp.cdc.gov/.../NHANES_<cycle>_MORT_2019_PUBLIC.d	6 files

About **204 MB**. Every URL, byte size and status is in
data/raw/_download_manifest.csv.

Note

Why keep a download that 404s? The lexicon was meant to classify medications by therapeutic class. It is gone, so the medication covariates fall back to questionnaire self-report. Deleting the failed download would hide that. **Three MISSING rows in the manifest are the honest record that a planned covariate was never built.**

Which component supplies what

Component	Key variables	Feeds
DEMO	RIDAGEYR RIAGENDR RIDRETH3 INDFMPIR WTINT2YR SDMVPSU SDMVSTRA	age, Asian cut, PIR
BMX	BMXBMI BMXWAIST BMXHT	FLI, adiposity, both group axes
TRIGLY	LBXTR LBDLDL WTSAF2YR	FLI, TG criterion, fasting frame
BIOPRO	LBXSGTSI (GGT) LBXSCR	FLI, eGFR
GLU GHB HDL	LBXGLU LBXGH LBDHDD	glycaemia + HDL criteria
BPX BPQ DIQ	BPXSY1-4 BPXDI1-4, BPQ050A BPQ100D, DIQ010/050/070	BP criterion, "treatment" arms
ALQ	ALQ130 ALQ120Q ALQ120U ALQ121	alcohol exclusion
MCQ ALB_CR SMQ PAQ	MCQ160* URDACT SMQ020/040 PAD680	comorbidities, CKD, Model 2
RXQ_RX	—	nothing. Downloaded, never read.

The merge, and the trap inside it

`read_component()` stacks a component across all six cycles and **pads any variable absent from a cycle with NA**. Then 16 `left_joins` onto DEMO by SEQN.

N after merge: 59,842. Safe here: SEQN is unique across all six cycles (verified, zero duplicates), and no joined component has more than one row per person.

Padding makes “never asked” look identical to “did not answer”

Variable	What it measures (CDC label)	In	So what
RIDRETH3	race/ethnicity with a non-Hispanic Asian category	G–J	Asian status unknowable for 2007–2010 → Step 5
DIQ070	takes diabetic pills to lower blood sugar	F–J	in E, “antidiabetic” collapses to insulin (DIQ050) alone
MCQ1600	ever told you	H–J	its % has a

Step 1 — Age eligibility

Step 1 — the four lines

***Paper:** “The study inclusion criteria were participants **aged >18 years** with evidence of MASLD.”*

Component → variable	DEMO → RIDAGEYR
Code	02_build_analytic.R:114, filter at :315
Decision	age >= 18
N	59,842 → 36,580 · ELIGIBILITY · 23,262 dropped

[!] The paper says $>$, the code says $\geq$

That is **70 people** — and it is not a PDF artefact. The published PDF contains **24 correctly encoded $\geq$ characters** ($BMI \geq 30$, $WtR \geq 0.6$, $FLI \geq 60$, $1 \leq PIR \leq 4$). Where the authors meant “at least”, they wrote it. At the age criterion they wrote a plain $>$.

Rule	N	Deaths	Groups I/II/III/IV	$\Sigma dev $
Paper	6,300	585	386 / 4,541 / 769 / 604	—
age $\geq$ 18 (ours)	6,371	586	391 / 4,585 / 787 / 608	71
age $>$ 18 (paper's words)	6,301	585	379 / 4,534 / 781 / 607	29

Counter-evidence: the paper's Table 1 mean age is 51.0 (16.6) — which is what ≥ 18 gives. Strict > 18 gives 51.4 (16.3). **This is a strong lead, not a settled answer.**

Step 2 — Steatosis via the FLI

Step 2 — the Fatty Liver Index

Paper: “A threshold of $FLI \geq 60$ was used to determine the presence of hepatic steatosis.”

$$FLI = \frac{\exp(L)}{1 + \exp(L)} \times 100, \quad L = 0.953 \cdot \ln(TG) + 0.139 \cdot BMI + 0.718 \cdot \ln(GGT) +$$

Input	Component → variable	Formula wants	NHANES gives
Triglycerides	TRIGLY → LBXTR	mg/dL	mg/dL ✓
BMI	BMX → BMXBMI	kg/m ²	kg/m ² ✓
GGT	BIOPRO → LBXSGTSI	U/L	U/L ✓
Waist	BMX → BMXWAIST	cm	cm ✓

N: 36,580 → **14,989** (computable) → **6,376** ($FLI \geq 60$). Median FLI 85.5 (73.7–94.7).

Reserved codes: there is no global rule

LBXTR — a lab measurement

77 is a triglyceride of 77 mg/dL.

Stripping 77/99 “to be safe” deletes **real data**.

ALQ130 — a questionnaire item

777 is *refused*, 999 is *don't know*.

Leaving them in turns 9 non-answers into people drinking **777 or 999 drinks a day**.

Important

Same-looking digits, opposite meanings — in the same dataset, in the same pipeline. **The codebook decides, variable by variable.** Check it before you compare a variable to a number.

[!] DESIGN is not missing data

Triglycerides come from the **fasting subsample**, which carries its own weight WTSAF2YR. The paper never says whether it restricted to that subsample.

- ▶ **21,618** of the 36,580 adults fall **outside the fasting-weight frame** — most were never sampled for it. That is **DESIGN**.
- ▶ **792** are inside the frame and still lack an input. That is **MISSINGNESS**.

Warning

Say “outside the frame”, not “never asked to fast” — the weight cannot certify that. **2,031** carry a weight of exactly 0, and **819** have all four FLI inputs with no valid weight at all, so they demonstrably *were* measured fasting.

Different problems, different remedies: a **weight and a population sentence** for the first; a **missing-data analysis** for the second. Never their sum.

Note

The arithmetic does not simply subtract: the full file loses **21,591** here, because the **819** adults with all four FLI inputs but no fasting weight are kept. $21,618 - 819 + 792 = 21,591$.

Two row sets, kept apart

Row set	FLI step	Final N	What it is
Full analytic file	14,989	6,371	every adult with all four FLI inputs. Saved as <code>masld_analytic.rds</code> , every table in output/ uses it
Locked domain	14,170	6,048	its subset carrying a valid WTSAF2YR

They differ by **323** records, and **neither is an error**.

- ▶ The paper does not report using survey weights → **6,371 is what reproduces it**
- ▶ A design-aware estimate is only possible on **6,048** → that is **P4's** job
- ▶ 6,048 is a *different denominator*, not a correction to 6,371

Warning

Step 3 — MASLD criteria

Step 3 — five criteria, at least one required

Paper: "MASLD was defined as the presence of hepatic steatosis, accompanied by at least one of the following cardiometabolic criteria..."

#	Paper	Variables	Our rule
i	BMI ≥ 25 or waist > 94 (M) / > 80 (F)	BMXBMI BMXWAIST	bmi ≥ 25 OR waist rule
ii	FPG ≥ 5.6 mmol/L or HbA1c $\geq 5.7\%$ or antidiabetic meds	LBXGLU LBXGH DIQ010/050/070	LBXGLU ≥ 100 OR LBXGH ≥ 5.7 OR dx OR tx
iii	BP $\geq 130/85$ or antihypertensive	BPXSY* BPXDI* BPQ050A	sbp ≥ 130 OR dbp ≥ 85 OR tx
iv	TG ≥ 150 mg/dL or lipid-lowering	LBXTR BPQ100D	LBXTR ≥ 150 OR tx
v	HDL ≤ 40 (M) / < 50 (F) or HDL-targeted treatment	LBDHDD	≤ 40 M, < 50 F — treatment arm not implementable

N: 6,376 $\rightarrow$ **6,374**. Two people.

Judgment call 1 — is the glucose cut 100 or 100.9?

The paper sets criterion (ii) in **mmol/L**. NHANES LBXGLU is in **mg/dL**. Someone has to convert, and the paper never shows the converted number.

1. Paper: fasting glucose $\geq$ **5.6 mmol/L**
2. Converted: $5.6 \times 18.016 =$ **100.9 mg/dL**
3. We coded $\geq$ **100** — the conventional rounding
4. **212 people** have a glucose between 100 and 100.9. Our rule counts them as meeting criterion (ii); a strict reading would not
5. So we re-ran MASLD with $\geq$ **100.9** and counted who falls out: **0 people**

The habit, not the answer

“It’s only 0.9 mg/dL, it can’t matter” is a **guess**. “I re-ran it and 0 people change status” is a **finding**. Same decision — but only one of them you can defend to a reviewer. Quantify first, *then* decide to leave it alone.

Judgment call 2 — and what Step 3 really did

A missing criterion counts as *not met*, not as *unknown*. `n_crit` uses `rowSums(..., na.rm = TRUE)`, so an NA contributes 0 — conservative: it can only keep someone **out** of MASLD, never let someone in. It also barely matters here:

Meet criterion (i), adiposity	6,329 of 6,371
Meet <i>exactly one</i> criterion	354
...of whom, that one criterion is adiposity	352
Dropped by this whole step	2 of 6,376

Why. FLI is built from BMI and waist, and criterion (i) *tests* BMI and waist — so selecting on $FLI \geq 60$ strongly **enriches** for criterion (i) without requiring it. (89 of the final 6,371 have $FLI \geq 60$ with $BMI < 25$; 42 fail criterion (i) outright.) It is near-automatic by enrichment, so the other four rarely decide anything.

Report a filter by its effect, not its definition. “MASLD required at least one cardiometabolic criterion” sounds like work; here it excluded **2 people**.

Step 4 — Alcohol

Step 4 — building the alcohol exclusion

***Paper:** “Individuals with significant alcohol consumption — defined as more than two drinks per day for men and more than one drink per day for women — were excluded.”*

The paper says it removed these people, so we built the rule. Getting it right needs two codebook checks that are easy to skip.

1 — “Drinks per day” is frequency × quantity, not ALQ130. ALQ130 is drinks per *drinking day*. Three drinks every Saturday gives $ALQ130 = 3$ but an average of **0.4/day**. Thresholding ALQ130 on its own removes about **2,000** people, far more than average intake would. So we rebuild it: $(ALQ130 \times days_per_year) / 365$, with ALQ120Q/ALQ120U supplying the frequency ($\times 52 / \times 12 / \times 1$), and ALQ121 replacing that pair in 2017-18.

2 — Strip 777 and 999 before comparing to any number. They mean *refused* and *don't know*. Left in, **9 people** here become 777- or 999-drinks-a-day heavy drinkers.

That yields a per-person flag, `sig_alcohol`. **427 people in the cohort exceed the paper's threshold.** Whether we then *drop* them is the next slide.

[!] We built it — and then did not apply it

`APPLY_ALCOHOL_EXCL <- FALSE`, so **no rows are dropped: 6,371 → 6,371**.

The flag is still computed, still saved into `masld_analytic.rds`, and the funnel still logs the setting — so the choice is *recorded*, not hidden, and flipping one line turns it on.

Why off: applying the paper's own stated exclusion moves us further from the paper.

Cohort	N	Deaths	$\Sigma \text{group dev} $
Paper	6,300	585	—
Ours, exclusion off	6,371	586	71
Ours, exclusion on (427 dropped)	5,944	546	356

- ▶ `reproduction_report.md` §6 says the +71 gap is “almost entirely the alcohol-exclusion ambiguity”. **The numbers do not support that** — Step 1's age rule fits far better
- ▶ So the paper's Methods and its own reported N **disagree with each other**. We cannot resolve that; we can document it, and default to the cohort that reproduces the tables

Step 5 — The four phenotypes

Step 5 — obesity × central adiposity

Paper: “Obesity was defined as $BMI \geq 30 \text{ kg/m}^2$ (≥ 25 for Asians). High central adiposity was defined as a $WHtR \geq 0.6$.”

	WHtR < 0.6	WHtR $\geq$ 0.6
Obese	I — 391	II — 4,585
Non-obese	III — 787	IV — 608

$whtr = BMXWAIST / BMXHT$ — both in cm, so the units cancel. Five people sit exactly on 0.6 and count as high, matching $\geq$.

The check that proves the construction is right

Our per-group BMI medians: **30.9 / 35.2 / 27.6 / 28.8** — identical to all four published values. That holds *independently* of the N gap.

[!] A cut-point you cannot apply to a third of the cohort

asian comes from RIDRETH3 == 6 — and RIDRETH3 **does not exist in 2007-08 or 2009-10**. The code recodes the resulting NA to 0, i.e. *not Asian*, for **2,322 of 6,371 rows (36%)**.

- ▶ Where the variable *does* exist, the Asian rule moves **121 of 250** Asian participants into the obese arm. **This rule has real bite.**
- ▶ In 2007–2010 the only ethnicity variable is RIDRETH1, whose category 5 is “Other Race — Including Multi-Racial”: it *contains* Asians but cannot isolate them
- ▶ 71 cohort members fall in it; only **17** have BMI in [25, 30) — the window where the rule changes anything

So: at most 17 of 6,371 are misgrouped. The paper states one rule for 2007–2018; the data support it only for 2011–2018.

The paper contradicts itself, and the data settle it

- ▶ **Methods text:** obesity = BMI $\geq$ 30
- ▶ **Table 1 and Table 2 column headers:** BMI $\geq$ 25 kg/m² / BMI < 25 kg/m²

The headers are wrong, and you can prove it without asking the authors: the paper's own group III and IV BMI medians are **27.6** and **28.8**, sitting in columns labelled "BMI < 25". Impossible.

Coding to **30** reproduces those medians exactly. Coding to 25 does not.

*Finding an internal contradiction and **resolving it against the data** — rather than picking whichever reading you prefer — is exactly what M2 asks for.*

Steps 6–8

Step 6 — covariates, where the paper says almost nothing

Paper: “The definitions of comorbidities in the NHANES study have been previously described.” — that is the entire specification.

Derived	Variables	Rule
htn	BPQ020, BPXSY* BPXDI*	self-report or $\geq 140/90$
t2dm	DIQ010, LBXGH	self-report or HbA1c ≥ 6.5
dyslip	BPQ080 BPQ100D, LBDLDL	self-report or meds or LDL ≥ 130
ckd	LBXSCR, URDACT/URXUMA/URXUCR	eGFR < 60 (CKD-EPI 2021) or ACR ≥ 30
pir_cat	INDFMPIR	<1 Low · 1–4 Middle · >4 High
current_smoke	SMQ020 SMQ040	current among ever-smokers

Warning

The code comment says these are “**calibrated to reproduce the paper’s Table 1 prevalences**”. That is fitting a definition to a target — **circular**. Report it as: *the paper did not specify, we chose X, here is the residual gap* (1–3 points).

Step 7 — the outcome is not in NHANES

Six fixed-width .dat files from NCHS, joined by SEQN.

Field	Columns	Used for
SEQN	1–6	join key
ELIGSTAT	15	1 = eligible for linkage
MORTSTAT	16	0 alive / 1 dead
UCOD_LEADING	17–19	001 heart ... 005 cerebrovascular
PERMTH_EXM	46–48	the clock — months from the MEC exam

Decision: PERMTH_EXM, not PERMTH_INT — the exposure is measured at the MEC exam, so follow-up starts when the exposure was measured. The paper does not say which it used.

Important

Verify the column positions before trusting anything. A one-column offset corrupts everything silently. Check: $\text{ELIGSTAT} \in \{1,2,3\}$, $\text{MORTSTAT} \in \{0,1\}$, $\text{UCOD_LEADING} \in 1\text{--}10$, $\text{PERMTH_EXM} \geq 0$, SEQN in the cycle's range.

[!] A trap inside our own saved dataset

Definition	CV deaths
<code>cv_death</code> in <code>masld_analytic.rds</code> — <code>ucod_leading</code> $\in \{1, 5\}$ Heart only — <code>ucod_leading == 1</code>	185
The paper	158

- ▶ `README.md` says “diseases-of-heart”; `reproduction_report.md` finding #3 says heart-only
- ▶ `04_table2_cox.R:42` quietly defines its own `cv_heart` and uses that, so the published Table 2 is correct

If you read the analytic file and use `cv_death`, you will disagree with every table in this repository. Use `mortstat == 1 & ucod_leading == 1` until it is fixed.

Step 8 — save, and count what you lost

```
s0 <- dat # 59,842
s1 <- filter(s0, age >= 18) # 36,580
s2 <- filter(s1, !is.na(fli)) # 14,989
s3 <- filter(s2, steatosis == 1) # 6,376
s4 <- filter(s3, masld == 1) # 6,374
s5 <- filter(s4, !is.na(group)) # 6,374
s6 <- filter(s5, eligstat == 1, !is.na(mortstat), !is.na(time_yr)) #
s7 <- if (APPLY_ALCOHOL_EXCL) filter(s6, sig_alcohol == 0) else s6 #
```

Two files: masld_analytic.rds (6,371 rows) and merged_all.rds (all 59,842, pre-filter, so any step can be re-derived without re-reading the .xpt files).

ELIGIBILITY · DESIGN · MISSINGNESS · TARGET POPULATION · OUTCOME ASCERTAINMENT — *if you cannot name a row's type, you do not yet know what the row means.*

What it all cost

The decisions ledger — where we may differ from the paper

Decision	Why	Cost
[!] Age ≥ 18 , paper writes > 18	matches the reported mean age	70 rows; the strongest lead on the N gap, not settled
[!] Alcohol exclusion off	applying it gives 5,944 vs 6,300	427 rows
[!] Asian cut only where RIDRETH3 exists	absent in cycles E, F	≤ 17 of 6,371 misgrouped
[!] <code>cv_death = heart + cerebrovascular</code>	—	185 vs the paper's 158
Both row sets kept (6,371 / 6,048)	unweighted paper vs design-aware estimate	323 rows differ
PERMTH_EXM as the clock	exposure measured at the MEC exam	6.76 y vs the paper's 6.9 y
Unweighted throughout	the paper reports no weights, strata or PSUs	the design-aware estimate is P4

The ledger — decisions the paper left to us

Decision	Why	Cost
FLI transcribed verbatim, no unit conversion	NHANES already supplies mg/dL, U/L, cm	none
No code-stripping in LBXTR/LBXSGTSI	77 there is a measurement	would delete real values
Strip 777/999 in ALQ130/ALQ120Q	there they <i>are</i> non-answers	9 non-answers saved
Alcohol = frequency $\times$ quantity	ALQ130 is per drinking-day	ALQ130 alone removes ~2,000, not 427
Glucose 100 mg/dL for 5.6 mmol/L	conventional equivalent	212 in the gap, 0 reclassified
Missing criterion = absent	conservative	negligible — 6,329/6,371 already meet (i)
NHANES BP averaging, not rowMeans	documented CDC rule	5,523 people's DBP changed
Composite comorbidities	paper cites prior work instead of specifying	1–3 pt gaps; calibration is circular

Did it reproduce?

	Reproduced	Published
Analytic N	6,371	~6,300
Groups I/II/III/IV	391 / 4,585 / 787 / 608	386 / 4,541 / 769 / 604
Deaths	586	585
Group BMI medians	30.9 / 35.2 / 27.6 / 28.8	identical
Group IV vs I HR, unadjusted → Model 2	15.14 → 2.91	15.13 → 2.89

Reproduced HRs: unweighted, full 6,371-row file (Model 2 on its complete cases). Published: the paper's Table 2.

Yes, closely — apart from a residual gap of 71 that is not yet resolved.
*Read crudely, Group IV has the poorest survival, and the paper reports that. Under Model 2, Groups III and IV have similar HRs against Group I (ours: 3.00 vs 2.91). The replication succeeds; the disagreement is about **which contrast supports the conclusion.***

For P1

1. **Work the loop, in order:** paper sentence → operational concept → NHANES variable → component → *that cycle's* codebook → raw .xpt → your code → cohort consequence
2. **Write four lines per step:** the quoted sentence, the component and variables, the decision and its reason, the N afterwards
3. **Type every funnel row.** DESIGN, MISSINGNESS and TARGET POPULATION are not interchangeable and do not add up
4. **Two decisions here are still open** (age, alcohol). Reproducing a number is not the goal — being able to say *why* yours differs is

Tip

Full detail, with every number and line reference:
reproduction/COHORT_BUILD_STEPS.md

One exercise

Open `data/derived/merged_all.rds` and answer, with code:

1. How many participants are **exactly 18**? What happens to N, deaths, and the four group sizes if you use `age > 18`?
2. How many people would the **alcohol exclusion** remove? Does applying it move you toward or away from the paper's 6,300?
3. Pick **one** variable from Step 0's availability table. Which cycles is it missing from, and what does the NA-padding do to the statistic it feeds?

Write four lines for each: **what the paper says** · **what you did** · **why** · **what it cost**.